

Exploring the Hard problems of mind and life; Conversations with Daniel Polani; Episode 10

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hello Daniel hi Adam how are you I'm doing okay how are you doing yeah not too bad uh I mean semesters are full flow so essentially everything has to pass through a bottleneck and that's me um not everything of course but all my duties basically I'm the bottleneck for them everything waits for me to process them and that sounds rough actually to have so much depend on a pinch point and where that pinch point is you well the problem is really that um it takes a conference resistance right so uh work is a bit like like a liquid it moves through the pipes and then finds a spot where um where it basically um is being can pass through and um I know that some people actually have this trick of actually dropping stuff or not doing stuff and so on and work automatically then gets re-rooted right so they don't get less work to do um but I can't do that huh so bottlenecks will in terms of I suppose like a like a historicity about them Paths of least resistance if there's some sort of topology describing the Dynamics of what's happening for any system you'll tend to get so where will you get like the maximal flows like where will you get like the or the the least resisted ones at least where do you get bottlenecks where do yeah that's interesting I mean of course um depends on on what you talk if you talk about material bottlenecks I mean a lot of that is known right there there's various um Min Cut Max cut algorithms where you say okay this is the critical path for the system uh if you cut that then the flow really collapses others are not so much so think of it as a with our blood vessels think of material flows think of for example War um War economies more economies you basically try to create a system that's not can cannot be easily disrupted okay and um living organisms are a lot like that I do think that's also part of the philosophy of of talents of different agility agenda so if you disrupt the system again and again and again the system blinds essentially to embrace the disruptions as a way to find out ah this is this is a weak point of my system okay and that means that you basically learn to avoid bottlenecks if possible however the system is left alone um and very very robust so you you know that the

system is going to be left alone and protected and everything you kind of what you can't afford to have bottlenecks you can you can bottlenecks are simply allow you specialization I mean look at for example the queen in a termite or ad Colony that's a bottleneck you kill the queen you kill the colony in a way I mean probably not in all species because then if you kill a queen another queen will be will be emerging but essentially the queen is a bottle the queen is the part through which the um generational turnover of that Colony flows um The Queen Is heavily protected so if you have a bottleneck then you essentially mean means you can afford the bottleneck because everything passes through that and you protect it you optimize it maybe you improve it and this is this is basically when you have bottleneck if you have a highly disrupted system system that has constantly to find ways to respond to disruption I would claim you will not get bottlenecks or at least not um prominent ones with the queen of a new Social in psychology I'm thinking of was it Anthony zader was talking about I think like um genetic inheritance as entailing a kind of like Auto encoding because of like you have to make it to the next Generation you have to pass through this dimensionality reducer um and so in this case like the queen of all reproduction of for the colonies contingent her she would be this bottleneck for inheritance but then she then makes for uh I guess both uh uh was it a highly controllable would you uh I think he said the conservation of misery and that she would create like a vulnerability for the system and that like you now have to expend resources to protect the queen um because if she's gone to college in that case this is this is related also to the bow tie um uh principle that uh John Doyle talked about I think I mentioned it in one of earlier sessions um I just realized that this is a very similar principle right um and I completely agree with the auto and Cooperative I I don't remember who where I saw that for the first time but I really do agree that one of the probably one of the reasons why the genetic bottleneck actually so effective is because it's gone from the auto encoding again and again and again and again and Link accounting codes only the part that needs to be Auto encoded the rest is you know you just grab it from the environment you do morphogenesis you do embryogenesis you you do all the concrete adaptations to the concrete environment and forget about everything else everything else is just sent through the bottom but of course the genetic bottleneck is not really so much a bottleneck because it's very distributed so there's something very interesting about it compared to the to the other one in the sense that yes there's a bottleneck in a way that you only send very little data to the next environment but this data is highly holographic so it's highly distributed through the population and through speciation it's distributed in space for individuals and it's distributed in quality or character because you have different species so um

so there is an interesting way this bottle like kind of evades their sensitivity of being a single bottleneck which can be easily attacked so instead of you don't have to protect the bottleneck anymore because there are bottlenecks yes but there are so many of and across many dimensions foreign insect Colony like so if the queen dies will there be some sort of like inhibition of like sexual maturation and maybe there would be like like a waiting uh waiting for that suppressive signal to be released and then someone else Rises to drinks and then suppresses others then you have the new one so you have some of that robustness built in by having uh uh a set of potential uh usurper Replacements you definitely have that right I I forgot the details now which which speeches actually exhibit I know for sure that naked moles have that so naked moles are actually state-forming mammals live underground and they have a queen and the queen is the only fertile female in The Colony and she essentially keeps the other females infertile by bullying them they're being basically bullied into um stressed out and they don't don't get hurt when the queen dies um then one of the others becomes a queen right so we know this this actually happens um we I'm relatively sure that things like that happen in certain bee colonies um that you might have a journal queen and the original Queen goes lost or that she leaves or there are different mechanisms but again I I'm not an expert in this field so I don't know exactly how it is but I completely agree right if a queen dies The Colony I mean the colony is not lost but of course it's costly for the colony Because unless it's in the cycle it means that you are essentially now um thrown back by some time to game resources so what would a few of a system so there's the extents to which your bottlenecks could be a unified concentrated or distributed and so in this distributed system you're saying there's like more robustness um but I guess well maybe harder to make this this whole system pivot because you'd have to get consensus among the different bottlenecks that might be harder not necessarily you get to can get specific speciation right so if you have multiple bottlenecks um for usually multiple the bottlenecks are linked to different locations different locations can be linked to different niches with different environments different uh selection pressures and so essentially you get out of each other's quick when they don't they are just separate by space then you have the problem of um fraternal conflicts because if I mean there's a the saying that says no no hatred and no Wars are worse than fraternal Wars they are the worst wars because they're very similar they're basically occupy the same each so basically the other one is a direct competitor to the niche you have it I mean that can work in positive ways if you have cooperation it can working on adversarial ways and when that happens then then you know it's not a good thing so it works best if the niche are either far away from each other

either distance or else in in um in um Niche relevance or Niche character so you don't interfere with each other like the the additive Exclusion Principle I think that might be the name for that I think it might be called the competitive exclusion principle where it's like yeah you're not going to have two species similarly adapted for the same Niche Niche coexisting like you'll have to have some sort of species speciation or Extinction like some one will have to dominate or not if like or something I mean we have seen that the 900 000 humans might might be I mean humans do have some land that's how uh DNA so if the separation has not happened to long ago there could be future in terms of I was wondering the different kinds of bottlenecks I'm thinking of both society as a mind in society of minds but um like so for instance like you know you could have like a governmental structure more like a monarchy or you can have something that's more like a Loosely Federated like assemblage of tribes or you could have for an individual like ongoing debate some you know around like what it means for a person to flourish where it's like the is the self um serving like a vital role like or as a controller or is it um uh basically uh suppressing the flexibility and diversity and um openness and diversity you would have from a person who is not integrated in that way according to like the coherent narrative which can become a kind of I guess I'm wondering about um the the the trade-offs the advantages and disadvantages of the kinds of bottlenecks for persons and societies like when do you so like what kind of like conflict could you have within itself without some sort of like rank in order of values or something like this well the bottlenecks are not emerging because uh somebody likes to have them right they are usually the expression of some kind of resource limitation okay so um uh it's not something you seek but it's something you find um as a as a society or organisms organisms do not seek bottlenecks but the point is organisms if left unchecked they will just exploit whatever is available to them at some point you end up in a situation where you are constrained you can't just do what you want you have a limited capacity and if you have limited capacity then um you end up having to arrange whatever you're doing into this limited capacity somehow this creates structure without constraints or to take up the topic of today's talk conversation um if you talk about bottlenecks that means you have limited capacity something is limited in capacity whatever it is energy or information or whatever it is and that forces structure upon the system now is it a good thing well from the perspective of the individual probably not right the individual would rather have the full freedom from the perspective of the system it creates complexity it creates that thing the conditions to scale up perhaps or to make it more robust or to make it more uh compatible with other systems or to have some kind of consistency across scales because otherwise you

you can't make the bottleneck work okay so the the point is it's like like always right if you're on your own you you make your own rules when you live with others and you can move away from each other you don't have to worry too much about rules but when if you live in confined space you have to find rules or your life will be held um and once you have rules these European rules can take very very different character depending on circumstances but also depending on History right even in the same circumstance you can have two very different histories um the probably the one of the best lab for such social experiments is popular in engineer where you can you have all these valleys there and sometimes the valleys are so well separated the tribes in that Valley will never visit the tribe in another Valley and uh they they may live not far from each other but they don't attract and it turns out not only has popular engineer a huge amount of languages so I I think I have the number in mind of 1200 languages but also extremely different cultures and we are talking here about the essentially same or genetically very similar population and they they immigrated to the island early on I mean the population might be as old as really 40 000 years old um they immigrated onto the island very early and and nevertheless they developed in completely different ways but inside there are Society they had these rules and these rules are actually quite characteristic so the rules themselves would act as order parameters but it seems like there's like a kind of I don't know if this is if it's like an entrainment relationship or maybe there's you could go they could go separate with complementarity but like when there are rules for a group of people it seems like this would require the individual to follow those norms to be governable by those rules there's almost like a chicken egg problem it seems where it's like it's like the rules create the like a kind of like or uh or a bootstrapping into being um I guess I'm kind of thinking of like the there's an example you're using um you mentioned earlier about like um I think it's like a longbow manufacturer requiring like basically like um there'd be people to to create the demand for it to use it but then once they're there and then you get a kind of positive feedback going but getting it started can be a little bit tricky it's uh yes I mean it really depends right once once you get it started it may be the discovery it may be the the economic economic niche that you need right so if you if uh both are probably not a good example because they are relatively easy to manufacture even very primitive ones and the advantage is immediately visible so let's take something what is it another example I mean a better example is probably or more and more an example we chose it more clearly is electricity or electrical electrical devices magnets things like that magnets are known for a long time I mean at least in China that we're known for a long time probably even even in other countries and the the Chinese

use them as compasses but only when it was discovered that okay they have an influence on the electricity okay that's one first thing and then that was not enough you needed also to see how to make this combination actually workable and then we lost you at um you're saying talk about compasses and then uh we needed to work problem once you have that you still have the problem that for example yeah I um the the transition to electricity so so you make the transition to magnets can influence electrical electrical devices and now the problem is you know we have electrical devices but at this stage it's still a toy right now when you learn how magnets can actually induce electricity you can build dynamos and now both Edison and and Westinghouse understood that to make dynamos viable you actually have to create a whole infrastructure for this to work right so you had to make the decision okay we have to distribute the interest it's not like I'm selling this product that people will just buy it they will not buy it without the infrastructure so you will not buy Dynamo if you can't deliver the electricity to everybody who needs it and this is here the the threshold is very high right so this is very um probably one reason why the Industrial Revolution was so difficult to achieve the Industrial Revolution was difficult to achieve because to make things viable a lot of logistical problems had to be solved um anything pre-industrial was much less depending on um large scale Logistics to be available right away I mean there were examples for large-scale Logistics which is of course the Bronze Age trading bronze is trading with big distances and you traded bronze or the materials to create problems from was a trade area um but let's put it this way this is something that can grow you can imagine that you know okay how to grow that and then you discover that you don't have enough tin or copper but others have and then then you scale up from there but there are things where infrastructure is really needed then then you have to really think about how to how to scale it up the Romans were also masters of scaling up there were masses of scaling up stuff I think you mentioned um it was like um during the last conversation about um like a scientific mindset and running experiments with kind of factuals and how this I think you're saying like it wasn't quite there even for the Greeks and the Romans they just got things done like they just go through some yes they were just really good at getting things done yeah yeah they were very practical they didn't care too much about Theory um it's very clear that the Romans Roman science was not if if you read plinyus I mean it's not it's nowhere close to what the Greeks were able to do right so um the the Romans were practical they've got things done and whatever was important to be done they were very very good at um so they had attention to detail definitely I mean their Engineers were amazing they're they're water Engineers for example when you get the water from from

dozens of kilometers distance you have to keep the the slope at a very low level so you don't get creative turbulence um and they've got fresh water in the cities I mean we did not have anything like that uh for centuries afterwards modern water system I don't know Byzantium I assume that it will have had inherited some of the Roman achievements but essentially it's a Roman achievement and after after the Roman Empire fell we essentially wait for the 19th century to have a proper UI system with a proper water water and delivery system yeah but but the Romans somehow scaled they they would I really found very interesting Sparta supporter was militarily very successful place but they couldn't scale the way they were set up they couldn't scale and um and the Romans could scale because they didn't care too much about people having to be pure Romans it would be okay okay we accept you as as Roman allies or auxiliaries and and they scale I always always joking that um the reason archaeologists like to dig out Roman ruins so much is because they know it's like like you know like a big fast food chain uh you go to a big fast food chain I'm not going to name any names from here somewhere and it will always offer the more or less say menu and more or less the same services and will look almost the same and the chairs will be the same and you're familiar and with Roman cities is the same thing no matter where you go uh how far away from home you are you always have it you often have a theater you often have firms you have avocado itself is always up look like like a like a franchise basically Rome franchise it's City model and they franchise their military and the French of the military yes unfortunately not well enough that that actually messed it up in the end but yes in principle yes were they the first to create the idea of I guess I'm thinking so so like a viking Shield wall seems analogous like a Roman Phalanx but this formation that if you hold it there's a hole that's where the cylinder's parts but you have to be in a certain mindset to like be able to be disciplined into holding that line but this is older this is older we know that the old Greek the Greeks did that already I mean the Greek chocolates uh basically or even the macedonians later right they had these Shield walls and you had to rely I think the shield was in the left the the spear was on the right and you had to rely on your right fellow to protect you so you trusted your the guy on the right with your life yeah if anybody ran away the the wall would break so you really relied everybody relied on then they right right does that have its origins in the Greeks or is that sort of warfare or an older idea because that I I think I think the reason the Greeks were this absolutely devastating military Machinery was this that was an innovation that was a Greek Innovation I think I don't think that the you know you have the battle carrier but that is kind of dating back probably to the hixels as far as I can um track it maybe maybe it's older but but I

think the Egyptians got it from the higgus um the didn't have it before the Persians had it in the battle against Alexander the Great but the battle Chariot was more psychological weapon but not wasn't very effective it was easily defeated right um and Alexander defeated it in the same way by the way as the Romans would defeat the elephants of Hannibal later it's exactly the same trick more or less right so so but the the battle Shield wall is actually uh Greek invention it gets revived in different ways later that tertio by the by the Spaniards and also which is less less well known the I don't know how it's called how do you call them The Westerns when they have the westerns and and um the Cowboys are being attacked and they build with a wagon wagons the building circling the wagons circling the wagons you know who invented that as far as I know the the Bohemians the later checks um it in fact it was probably invented during the site were Wars wars of the precise which is a Proto proto-protestant um religious group which um thought of the the rulers and it was a very today you would call the Civil War essentially and they didn't developed that also as a kind of battle strategy but it was super efficient super effective and circling the wagons is very similar to some extent to the to the shield walls but super effective if and only if the you don't have too many people defecting on creating that formation otherwise it unravels yeah but it didn't I mean these people fought very well I mean I think it was invented by yamchichko was um the military leader and he he was blind as far as I know and he he was never defeated I guess I'm wondering about they're not necessarily incompatible but they see they feel that way at the moment um two different way or at least two different ways of getting let's say three I'm getting that kind of sensibility so like one would be of like you actually are playing out the Richer rollouts into the future and you can actually see the consequences of like the construction forming or not and so it's like it's almost like rational self-interest with with imagination uh just another one of like losing your like having creating a coast self with the other it's like one's more like planning and the other is like more of like an immediate coupling of like we are one body and who says it's mutually exclusive I don't know it feels different but I don't know why it is different it is different I agree it's very different today but they could actually feed on each other right so you could imagine that there are these two two drivers and they happen to join for a joint root this is something that I very much um very much think uh we totally underestimate right we always say oh this is the agent and so on but um Michael lachman has this idea that actually individuals um are not just individuals um they basically focus groups of traits or focus groups of call it memes or genes or or entities that that are kind of working together for a limited time when the individual dies then they dissipate or they pass

in case of genes or in case of memes they pass on to other individuals but the individuals are basically just vessels of these traits and so I would not be loath to say yeah there are two effects here Mutual self-interest based on future production or simply [Music] the solving the self because simply everybody shares the same meaning with you which is what for example religions do that's uh I mean one of the main arguments of Harari right so religions is a kind of synchronizing element but of course you could also say you have one story or more or less one narrative and this narrative is holographically encoded in many entities and that happens to make the entities collaborate not because they want to because they just simply now synchronize as per nature naturally synchronous as consequence of that but for them to synchronize I suppose so there's some synchronization manifold describing the coordination that will happen where individuals might form a larger super individual for some purpose so what conserves a myth potentially can serve this sort of mechanism design function is the idea of Truth one of those myths that we need but is also very dangerous like do we need to have some sense of an overarching narrative to create some kind of larger individuality well you can live with different truths and people have done this so right so when people had all these different peoples and say say in the Fertile Crescent they had all these different gods and there was this goal for that and that's good for that and they were not identical they were they were slightly different there were differences they would fight sometimes they would not even acknowledge their own the mutual existence they were just there but but they would not actually consider them part of the part of their their theologuni sometimes they would be absorbed we know that for example Hermes and hamstrings megastos hemispheres Mega stars as actually supposedly Greek but actually it's a Egyptian god deity and it was basically absorbed into the into the Greek uh Greek um Pantheon the Romans famously um identified their gods with a corresponding Greek gods and the identification is not always sometimes it fits Jupiter and and Jews yes they can be fit and maybe they are even the same deity in the Indo-European context but there are other deities which don't fit and they don't you you kind of tack them together but it's not really not really well founded and there is for example one one date in Roman society that the Greeks don't even have and that yet that date is more important to some extent than Jupiter itself himself and that's Terminus permanence is the god of the borders so if you have land borders or you have a piece of land and you put put the borders that isn't there is a deity there there's Romans have that they already have claims of stakes this is so very unique where do they get terminals do they invent him or borrow well I don't know where the where it comes from but it is uh Rome deity I don't think the Greeks

have anything analog nothing like that it's very clear that the Romans had the concept of land possession much stronger than the Greeks had yeah and um to correspond to it that is fascinating yeah and the Divinity is quite quite high up in the in the ranking I think I think he's ranking with Jupiter either together or above him or or I think he's very highly ranked there are no mythologies about them which about Terminus 2 which is kind of weird because it means that um you don't joke about land position right it's like you don't joke about security when you pass past Security in the airport you don't make jokes there this is not something to joke about the so truths what deities can be very you know basically overlapping and not not really messing up but the idea that there is some coherent story emerges of course in monotheism uh of course it also emerges in the view of the Greeks of the cosmos of the universal order of the world and in fact what does the change it changes a couple of things in the moment you say there is this kind of joint I'll avoid the word Truth for a second join reality it's a joint thing that basically decide this is what we all have in common several things happen number one it means we all play on the same playground yeah we're not like having basic playgrounds that interact and don't actually don't actually um intersect um there are actually some horror stories which live on the premise that you have actually multiple worlds that they kind of intersect and only sometimes they they meet and then then you see spirits and ghosts on the other side there's a fantastic uh Horror Story by Elgin on Blackwood the the Willows if you don't know it it's it's it's a really chilling story absolutely chilling story uh very well written backwards I think that's the name yes otherwise this person was faded to do what they're doing yeah no I think I think that this is considered a classic The Willows is considered one of the classic horror stories right on the level of Dracula and and things like that or or the or the or the monkey's paw or something it's really classic and it's chilling it's really I don't recommend doing reading it in a camping tour really not don't read it I can't think to it the best best to read it when it's lots of people around you and uh it's it's bright light daylight the cafe or something that's probably okay but otherwise really anyway so the the realities multiple realities suddenly are not multiple realities anymore they live in the same reality which is forces several things it forces order it forces okay there must be common order to everybody we all have to agree on certain principles that are valid for anyone and not just you can't just choose mix and match you can't just ignore each other in other words in the language that we used before you have constrained in the mental space of what's acceptable there's a mental space and what you move and you're basically sharing the space right so instead of having an airplane all by yourself where you can just you know um lean back and just spread yourself on the seats and just

ignore what what's going on around you you're now sitting very tight with the neighbors in the same spot in the same airplane and you have to okay we'll share the we should we'll share their the arm rest and I will not lean back too much and and you basically start having to negotiate your way around um on the other hand um well we do believe when we do science that the reality is like that it's a shared reality right so um you can't just argue reality away and it's a very interesting story can you argue reality way at some to some extent you can you can you know you can get through life very well being a flight earther perfectly fine right and as long as you do not have to um do um navigation on the on large distances you can be a flat earther and live your life without any problems not that I'm defending that as a scientific statement but I'm saying you can decide I'm a flat earther and that is fine and that that's perfectly well um you can say for example you can you know if you live in the US you know exactly what I'm talking about when I talk about imperial system right you can live without the metric system it's perfectly okay to do that um in fact well one of them it was okay until uh um a couple catastrophes with NASA where they were having trouble keeping track of units wasn't always there were some externalities that because they had to talk and then I think the people who did the used imperial system were not the Americans it was the British I think it was a British that did that in in the moment you have to talk to others you have to have a joint reality you have to have a joint reality in which you operate otherwise you know it doesn't work but you you have a point okay so one okay how long is it convenient to work with um with imperial system so if you're a medieval person and all you do you measure distances with inches feet yards miles these are related by integers integer scalings so I think it's 12 inches is one one foot three feet is one yard Etc so these are integer numbers they're not all consistent but there's always an integer factor between them same if you take a liquid arms as a weight or a liquid ounce as a volume or a liquid or an ounces of weight they are related by water as a weight of water so it's a volume um area is measured again differently but the point is and this is very important and as long as you do not mix areas volumes um lengths and these things with each other you do not have the need to unify anything you can talk about engines you can talk about uh ounces you can talk about Stones okay ounces and stones are related because the weight and volume can be translated into each other very easily but otherwise you don't care about it only in the moment when you know our Force you do mathematics you start doing integration measuring the volume of of objects using lengths that's basically Kepler has a very nice uh very nice little uh piece of work where he um essentially derives at integration rule for the volume of barrels says okay there's a barrel and you want to know how much volume is in the barrel by three

measurements and he then rewrites the volume of the snap Barrel just by these three length measurements so now there's an integration the moment you start doing that you are forced to convert length into volume measures the metric system makes sense although even then like it seems like what you're describing it's better than the truth in that it's like basically a capacity for ensuring coherence where it's needed and harmonization among like modes of sense making and adaptation but like something like a I guess maybe better than the truth something like um a semi-sacralized epistemic Commons something like science like the way we make like like our Notions about like uh was that simply it's a joint illusion it's a joint illusion if I have the illusion um like you have the same illusion we can agree on the illusion and then we don't don't fight about it and now we'll talk about let's talk physical truth which is probably easiest to have physical truths um first of all I don't like the word at all right so not that I don't like it I think that it's a dangerous word because whenever somebody talks about truth um I get edgy I see red flags all over the place um if somebody says they know the truth but that that's edgy because even as scientists we don't know the truth what we do is we try to approach an abstract truth what doesn't mean we're approach out that's truth we try to agree on what will happen in the physical world if we do certain things and we try to make find the reaction of the real of the physical world to what we do we try to find things that make us all agree on what's going on there so it's a mixture of getting the physical world to do certain things and us having a model that as transferable and that everybody agrees upon as well as possible so that if I do for example let's take a simple example if I throw drop a stone the stone drops and um then there's a acceleration you know t^2 is how much how much it falls per per after a given time and you do the same experiment you get the same result um to the Precision that you have then we kind of agree this is this is kind of a modern model of what happens but of course um then there are refinements this and that and the various things that are very specific to your place perhaps the gravity is slightly different perhaps you are out in space when things again work differently and then not you have to find a theory that explains that so essentially the truth doesn't actually exist out there there's something happening there's some physical Machinery that does stuff we um a theory that kind of through which we look at what happens out there and if that theory is consistent across individuals that do the same experiment then we say okay this is scientifically valid Theory and now we could say well this is the scientific truth but it's not the truth because it could be there are some effects that we have not described with this Theory and we haven't seen them or the theory is inconsistent and we have failed to see that it's inconsistent or the theory is okay on certain time scales but not on others Etc et

cetera so the concept of this is a scientific scientific truth is a very dangerous concept right um as a you know as a simple approximation it's okay to say oh this is the truth scientific truth but it's a dangerous word to use because people Pretend This is that this is the absolute absolutely Undisputed thing that's going on and that's usually not the case it gets worse when you start going to things softer topics like um um in biology it has improved but for example in medical medicine it's already getting difficult what's the truth what's the illness of this person um what do they react to the different people react to different things in the same illness right you go to history it gets even worse right so was an evil emperor was he just maligned by his contemporaries and the problem is now everything we know about him was filtered through his contemporaries so how do we approach something that if we were to live in Roman times and would see what Nero is doing would be consistent with what we think about I've been wondering about so it seems like this truth with a capital T this could you call it like a kind of neoplatonism and like it's like a little bit more of a Greek notion but like the Romans they would have more of this like pragmatic truth but like they were getting things done and so they would have basically the truth is like the shelling point of reality which is like they're coordinating around what actually gets these done like I say it's like this you say it's similar but and can we come to like functional agreement but this like truth to which one would uh bend a knee and shut down judgment uh or not shut down judgment but like um a singular type of truth that that seems I don't know is that the Greeks the problem is maybe it exists right maybe there's some simple the simple truth that actually exists um the problem is we will never know we cannot never we can not ever know um and the the the the problem is that this is often used as um you know for example political argumentation um because it creates like this Direct hotline to the deity direct hotline the data has told me this is the truth when that happens and I just so happen to have the hotline like yes and and yes and people who has the phone yeah yeah I mean um let's put it this way some people may have had the hotline I mean Newton and Gauss and Einstein they probably had a hotline right or for Neumann um it's very hard to believe they didn't but but you know this is this is very very rare that's that's it uh I I always like I'm both very compelled by it and it sits a little uneasy with me like but Nietzsche's ad hominem against Socrates where he says like it was like a um a Beauty value in culture and out of a kind of like resentment and a kind of power grab he created this world of philosopher Kings and ideal forms where he could like where his will to power could be satisfied but like I don't know how far I want to go that makes me nervous but this idea though of like something tyranny prone uh and and and a kind of uh even like a domination impulse

being like caught up within it of what would seem to be just like oh truth what could be truth goodness Beauty careful yeah yeah I mean let's put it this way I accept it as a as a working hypothesis I accept it as a tool to guide thinking I accept truth as a tool to direct where do we want to go as a scientist um scientists often say we're looking for the truth and I don't think they mean it in a negative way what they mean is they try to find this ground on which everybody can agree um that things can be interpreted jointly interpreted and can be jointly predicted I would say this is what I would think is what we understand the scientific truth very pragmatic right it's it's we can jointly interpret something and we can join the prediction so far I can live with this what is dangerous is that if that turns out to be ill-defined and we have seen that very well actually in the covet crisis the covet crisis one of the big problems that science was facing is Sciences scientists are usually quite well aware of the uncertainty of measurement in topics like physics um the uncertainty has been essentially almost squeezed out of the system I mean it's always there right but but it's so far away from anything that's um tangible that people can say oh gravity gravitational constant at this location is g equals blah blah blah so so many digits you can all say State them that's almost as good as truth as long as you don't do cosmology or something right As you move towards biology As you move towards medicine Things become very murky the statistics is really bad right and one reason it's bad is not just because statistics is is hard to do but because of that that's my personal opinion though biology doesn't want to be statistical in the sense as an organism you want to be unpredictable you don't want to be easily clusterable because if you are easily clusterable some viruses can find your weak points in fact if organisms didn't have to procreate it and find a find a partner to mate they probably would each of them trying to go in there their separate Niche and really be different from others incomparable of course that's not possible because they have to you know coordinate if they're made they have to communicate there are all these pressures to try to keep them together but if they could choose they would be very different right choose evolutionarily choose um that's a really interesting tension so there's a slight tension for regularity and legibility from coordinating in certain ways but if you optimize for that too much you are now a Sitting Duck for another uh of an antagonistic Optimizer exploiting you the conservation of misery strikes again exactly yeah you're basically opening the door to a vector it's a bit like having all your money in One Bank right it's again John Doyle's um John Doyle's um uh bow tie idea in a way in an abstract way so what did what what does it mean to us it means that if that's true and even if it's not true but but we know that the statistics and biology is not good and so there's a lot of uncertainty with

every statement that people made about covid is going to do this and it's going to kill that many people and on this age and this has that effect it's incredibly difficult to disentangle um scientists accept that and they realize that and they have different opinions what should be done because the problem with doing is that doing is a bottleneck again right so there's a stuff going on but when you have to make a decision the decisions are not arbitrary you can't just create some kind of big complicated uh super action decisions actions that you take on a political level or an individual level are always low bandwidth processes so you better invest the bandwidth well so for example if I say I'm going to do a lockdown now why I'm going to do the lockdown in three months that's a big decision you can't just undo it this is a decision you make now or later or you can postpone it but this decision has every decision is low bandwidth but very high impact the general public has a problem understanding that because they are used to clear facts and clear decisions usually so when I'm driving a car I go left I go right it's not always right there the prices situations where you don't know what how to react but most of the time decisions are simple simple and clear and and you do this way or do this way there's a reason why many companies have all these guidelines and uprs and and and rules and regulations so that you don't have to make decisions the decisions are essentially made for you because this is a condensed crystallized um crystallized experience or um institutional knowledge in covid basically we were caught outside of the Playbook there was no playbook for covid nobody knew what to do it's entirely true actually there were several countries including UK that actually had a contingency plan but it was not enacted so the contingency plan was not actually carried out but this is this is the only secondary the important part is that the general Republic was very confused and then people said oh let's do this let's do that let's do this let's do that or the scientists say that no the scientists say this so the problem is what's the truth now what is the truth now the truth or the approximation of the truth would be you create a million clones of planet Earth and start them at the same point in time say December 2019 and let them run and you try out all these things and get the statistics then you know what is the best action to take this is basically what you would do in reinforcement the problem is you can't do that if you don't have that model you don't have the opportunity you can't just restart the simulation and you don't know which factors end up actually affecting the outcomes if you look at Italy Italy had a huge number of covid deaths at the beginning in bagamo which is a rich area so that you can't say that these people don't know don't have a health system they have a good health system but but I I don't really know exactly why why they died and then for example the UK initially did not have at all any

coverage of this out there fine and then a month later boom it it exploded so the question is what are the right things to look at and different people made different guesses and people got very angry that politicians looked so helpless but the reality was maybe politicians should just have I mean sometimes politicians said we'll do that has made a decision like in New Zealand they decide okay hard luck though that's what they did hard knock down they can't covet out until Omicron right when Omicron came it was they had to give up it was not possible anymore but basically until Army Quran they had coveted under control um similar what happened to Spanish Flu in Samoa you had Western Samoa and you had American Samoa American Western Samoa I think had the death rate something between 20 or 30 percent in American Samoa they locked down everything strictly and they had no deaths of the Spanish Flu and these are neighbors okay the concept of Truth here is much more wobbly because nobody can say what was the right thing to do we have a computer to do the rollouts and we didn't have the and the credit assignment problem was too intractable even if we could and he didn't have the data we didn't we did not know um what what other parameters is this the contagion by um people meeting up is it is it a school was it um touching stuff was it air I mean we know now that it's mostly airborne right so the whole disinfection stuff was probably Overkill but who knew nobody knew that at that time so people say oh we should have not no we did not know that but it's like we need uh something like like maybe like a lesser Roman god like Terminus might be too much for truth but like I don't know like what's like a veritas yeah yeah I think I think the problem here is really we need to be a little bit more modest in our expectations um we need to accept sometimes you do not know sometimes there is an uncertainty and that is part of the game it is not resolvable and and then you have to ask okay what's the best I can do with my knowledge and that opinion can differ especially if there are different levels of risk aversions so some people are very risk-averse and other people are not and the question is who's right maybe none of them I mean it's you climb mountains you like climb climbing mountains you know and then free climbing it's dangerous but you accept the danger um you tell me to do that I say no I'm not going to do that I'm not this is this is not my level of risk but that's fine the problem with covet is you cannot be risk friendly on your own when you are risk friendly you involve others and now the question is where's the balance what's balance to strike the flat earners Earth areas there's definitely negative externalities there for beliefs like the flat earners earthers it's it could be just fine like maybe it doesn't hurt anybody I mean they don't have to travel to Australia nobody forces them to travel to Australia like here there's there's failings as we need to maintain to we need to hold the line and we need

we need the epistemics to be conducive to that otherwise you don't it doesn't work you don't achieve the the problem with with the covet is actually really interesting because I think that when you talk about um I had I had conversations with various people and they were the people that for example didn't believe in vaccination didn't believe in masks and so on and so on now the point is you know as long as it just affects them it's fine right why not the problem is I remember very distantly when with the whole covet thing started that people working for Trump's transport of London people who work for for the Public Services the the buses and so on they're I think Sev dozens of them died in a very short period dozens of and and passengers for example casted at drivers I mean seriously this this is clearly it's very clearly not neutral right it has an effect people are affected it's not it's not like you can say okay it doesn't matter people were affected by this there were other things like um nurses and and I talked to I talked to to to um to people in the in the health in the health department basically they just say I mean people just ignore things or or claim it's not happening and it is happening I mean these people work with that now they're they're of course nurses say I don't want to be vaccinated I can I can see the fear of vaccination some people are sensitive to that some people may not not feel um feel that their their body doesn't react well to vaccinations in general you know I can't see that one The Mask thing is a different story sometimes it felt really like I don't wear the mask because this is a political statement not because just it's uncomfortable it's a political statement it's almost like smoking his sign no smoking they smoke right at some point smoking just became obsolete people just it was enough you didn't do that you wouldn't it just it's not it's not doesn't make you cool it's like uh seat belts and cars seat belts cars at some point there was a really big discussion shall we do that shall we do seat belts kill people sometimes yes they do still they save many more people than they kill and that that is kind of the the type of thinking that at some point basically people just tire off the argument that's that's what happens so it's not like they they converge to the truth they just tire off the argument now if today I will just invent a case in a very important paper there is a study that proves no seat belts actually kill people more than they help them you know then people say oh 50 years we did the wrong thing right but this is how knowledge works I don't think this is the case by the way I have to you know disclaimer yeah I do believe seat belts really help but I'm just saying that if we find there's a reversal of our understanding it's not a reversal of Truth it's not that we were lied to either it was just a collective of the best knowledge we had at the time that made us take action and sometimes you can refine sometimes you find okay perhaps vaccination helped maybe we shouldn't give them to people who have these these these illnesses maybe the mask did help

didn't help we don't know and some people say well the mental illnesses has increased during lockdown so the lockdown was a bad idea maybe the lockdown should have been earlier and then it could have been spared all together for example if you had done blocked on at the beginning of Corona squashed Corona right away nipped it in but maybe we wouldn't have the pandemic in the first place or we would have had vaccinations so early that it would never have gone to that level that it went so we made the concept of like truth a very task that's that's strong enough that we can coordinate around it but not one that becomes basically uh inflexible and tyrannical and resistant to updating exactly and so it's so not quite yeah it's Carl Sagan's book um uh was it um a demon Haunted World science is a candle in the dark that um gave me a I think that that replaced religion for me when I was young and it gave me like uh there was like enough of a of a sense of like even like a Holiness about the Endeavor but like there's a beauty to it but it was imperfect it was like you're always wrong you're never you know you're you're always gonna you know never assume you're not being fooled in some ways and but this like but it's still precious like you're still seeking to do the best you can yes that's interesting but it will always elude you it's just a candle you know it's and it is always wrong I I also think it's probably I know that Papa meant well but I'm actually thinking his way of of saying things is actually too negative and all saying always wrong is also too negative um because it implies oh the scientists don't know anything that's not entirely true people live much longer today uh you can do operations on people um and save them which would have died um 100 only 100 years ago you can heal things that aren't unbelievable you can extend people's lives by by a lot and really unbelievable circumstance we can fly to the moon so I don't think we are wrong wrong is not not the term I would choose I would say something else we should accept that's an impressive candle that can take you to the Moon yes that's a big one yes 110 meters as well yeah and the candle burns from the bottom very important it is the decision that we're trying to take should be reflecting our very best approximation to this jointly consistent grounds that hold up as well as possible to the Future doesn't guarantee it's a case but this is what we want when people say do something because of an agenda it means that they're not caring about this they're caring about creating something favorable for them they take basically they are partial they they are partisan they are partisan and they decide okay we have picked this direction um for our purpose this is what the say people on the street are afraid of being taken advantage of I don't blame them for that right people are very afraid of being taken advantage of right sometimes um they are right evil it's probably true um in enough or true it's probably a good good way of describing the situation in

enough cases that they're probably right still I think we need to keep not just the camel we need to keep a flag of Truth seeking brother truth truth seeking up saying we are trying to find the what's going on we're trying to be consistent we're trying to do decisions that are reflecting your best understanding of what what's a good future but of course agents have probably different thinking of what's good for the future but that's a different story that in my opinion is not part of the story of truth anymore that is part of of the story of politics and let's get gets nasty when politics distorts truth in the service of partisan agendas thank you Daniel thank you I'm looking forward to next time